

AURORA FILTER

Retrofuturist Audio Processing Unit

Design Specification v1.0

DESIGN PHILOSOPHY

Aurora Filter is not vintage audio hardware.

It is a vision of the future from 1974.

The plugin should feel like laboratory equipment from an alternate timeline where Braun, NASA Mission Control, Olivetti and Bang & Olufsen collaborated to build audio tools for a lunar research station.

The design language is:

- Clean
- Functional
- Scientific
- Confident
- Warmly industrial

Avoid:

- RGB lighting
- Cyberpunk aesthetics
- Skeuomorphic metal textures
- Fake rack ears
- Excessive shadows
- Modern glassmorphism

The UI should feel engineered rather than designed.

COLOR PALETTE

Base Surface: Dirty White

E6E0D2

Secondary Surface: Warm Beige

D8D1C1

Panel Borders:

7A7468

Primary Text:

1F1F1F

Secondary Text:

5E594E

Accent Amber:

D08A2E

Accent Mustard:

B8943A

Warning Orange:

C56A2B

Graph Background:

171717

Graph Lines:

FOB547

Indicator LEDs:

D99A33

TYPOGRAPHY

Headings: Eurostile Extended

Body: IBM Plex Mono

Numerical Readouts: IBM Plex Mono

Characteristics:

- Uppercase labels
 - Generous spacing
 - Technical terminology
 - Instrument panel feeling
-

WINDOW SIZE

Default:

1400 x 900

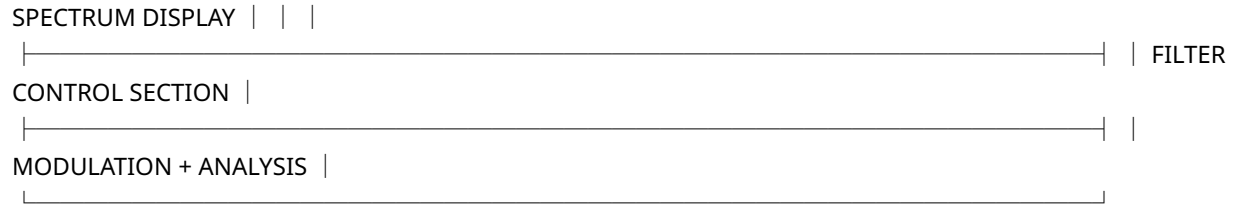
Resizable:

Minimum: 1100 x 700

Maximum: 4K

OVERALL LAYOUT





The spectrum display should dominate the interface.

Everything else supports it.

HEADER

Left:

AURORA FILTER MODEL AF-74 MK III

Center:

Preset Selector

[Previous] PRESET NAME [Next]

Right:

A/B Oversampling Settings

Power Button

MAIN DISPLAY

Height: 40% of interface

Black display area

CRT-inspired appearance without fake scanlines.

Contains:

- Live spectrum analyzer
- Frequency response curve
- Filter node visualization

- Gain response

Graph Color:

Amber

Grid:

Subtle dark grey

Frequency Labels:

20Hz 50Hz 100Hz 200Hz 500Hz 1k 2k 5k 10k 20k

FILTER SECTION

Central module.

Large panel.

Contains:

Filter Type

[LP] [HP] [BP] [NOTCH] [PEAK] [SHELF]

Selectable buttons.

PRIMARY KNOBS

Large retrofuturist knobs.

Diameter: 70px

Dirty cream plastic.

Black indicator line.

Controls:

Frequency

20Hz – 20kHz

Resonance

0.1 – 24

Drive

0 – 24dB

Mix

0 – 100%

Output

-24 to +24 dB

NUMERICAL READOUTS

Every knob has a dedicated readout.

Example:

FREQ 824 Hz

RES 3.2

DRIVE 4.1 dB

MIX 100%

OUTPUT 0.0 dB

FILTER VISUALIZATION

Moving node on graph.

User drags directly on graph.

Node controls:

Horizontal: Frequency

Vertical: Resonance

Mouse wheel: Drive

Double click: Reset

MODULATION SECTION

Bottom Left

Panel Title:

MODULATION

Contains:

LFO Source

OFF LFO A LFO B ENV

Destination

Frequency Resonance Drive

Amount

-100% → +100%

LFO ENGINE

Bottom Center

Mini oscilloscope display.

Waveforms:

Sine Triangle Square Random

Controls:

Rate Depth Phase

ENVELOPE FOLLOWER

Bottom Right

Input level visualization.

Controls:

Attack Release Sensitivity

Display:

Small amber waveform.

FOOTER

SYSTEM STATUS

Sample Rate

96.0 kHz

Oversampling

4x

Latency

0.23 ms

CPU

1.7%

Signal Path

MONO STEREO MID/SIDE

INTERACTION DESIGN

Animations:

Minimal

Only:

- Knob rotation
- Spectrum updates
- Meter updates
- LED indicators

No bounce.

No elastic motion.

No glowing effects.

VISUAL REFERENCE

Imagine:

Braun ET66 calculator

+ NASA Apollo control systems

+ Olivetti industrial design

+ 1970s scientific instrumentation

+ Apple-level layout discipline

Result:

A premium retrofuturist filter plugin that feels like a precision instrument from an optimistic future that never happened.